

Noodle Shop

Restaurant ordering with authentication and natural language AI

WEB PROGRAMMING PROJECT

Django · SQLite · OpenRouter API

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Project brief

A restaurant website with two ordering paths: browse the menu or describe a request in everyday language.

Customer experience

Browse dishes, adjust a cart, and submit an order from the website UI.

Project goal

Connect authentication, access rules, order management, and an AI menu assistant in one Django application.

Customer authentication

Register

Enter an email and password, confirm the password, create an account, and start a signed-in session.

Sign in

Authenticate with the registered email and password. Django keeps the signed-in state in a session.

Sign out

End the customer session through a POST request from the website.

Account record

Registration also creates a member points record, ready for reward features.

Authorization and user roles

Guest

Browse the menu, use AI ordering, and place standard orders. Rewards and points require an account.

Member

Use account-only points and reward features. Access to a receipt is tied to the signed-in user.

Staff

Manage orders and application data through Django Admin with staff permissions.

Access is checked on the server. The interface also adjusts links and reward controls based on sign-in state.

Application architecture

CUSTOMER BROWSER → **DJANGO VIEWS** → **SQLITE**

HTML templates + JavaScript | Python request handling | Django models

AI ORDER REQUEST → **DJANGO SERVER** → **OPENROUTER**

CSRF protected JSON POST | API key stays server-side | Chat completion response

The server validates data before returning menu items to the cart.

Natural language food ordering

1 Customer types a request

“Two chicken noodles, thin noodles, extra egg”

2 Django sends the menu and instructions

3 The model returns structured JSON

4 Django checks every field; the browser adds valid items to the cart.

EXAMPLE RESPONSE

```
{
  "product_id": 5,
  "quantity": 2,
  "noodle": "Thin",
  "egg": true
}
```

Unclear requests return a follow-up question. Recommendations do not change the cart.

Orders and stored data

Customer account

User stores Django account details.
UserPoints stores a one-to-one member points balance.

Order

Payment method, status, subtotal, tax, total, and timestamp.

Order items

Each order can contain dishes with quantity, price, noodle option, egg option, and reward status.

Checkout

Server-side catalog prices are used when the order is saved. A receipt is shown after successful checkout.

Validation and testing

Request safeguards

CSRF token, POST-only AI endpoint, message length check, server-side API key, and validated item fields.

Order safeguards

The server checks menu IDs and quantities and calculates prices from its own catalog.

Automated checks

Tests cover registration, sign-in and sign-out, protected points, rewards, checkout, and mocked AI responses.

Before deployment

Set a unique Django secret, turn DEBUG off, and configure allowed hosts for the deployment environment.

Submission materials

GitHub repository

github.com/Adenan1305/django-project

Presentation slides

submission/Noodle_Shop_Presentation_Slides.pptx

Article report

submission/Article_Report_English.docx